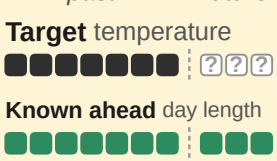
How Toto-1.0 makes a forecast

Datadog · 151M parameters · decoder-only transformer, time and row attention · zero-shot

1. One row per series

Targets form one group of rows; known-ahead covariates join as extra rows. A ? is a value not known yet.

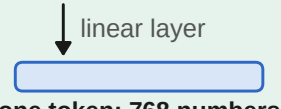


Past-only covariates such as the NAO index are not used.

2. Scale as you go, patches of 64

Each patch is scaled with the mean and standard deviation of the data up to it, so no future value leaks in.

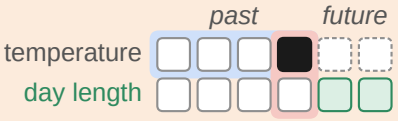
$$\tilde{x} = \frac{x - \mu_{\le k}}{\sigma_{\le k}}$$



one token: 768 numbers

3. Eleven time blocks, then a row block

The last context token (black) predicts the next patch. RMSNorm in front, shortcut around.



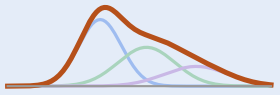
Time attention: causal, along the row (RoPE + xPos); 11 blocks

Row attention: the 12th block, down the column across the group

Feed forward: SwiGLU, 768 → 3,072 → 768

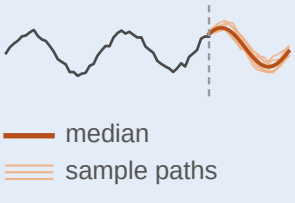
4. Sample 64 steps at a time

The last token gives a **mix of 24 Student-t curves** for each of the next 64 steps.



one step: a Student-t mix

A sample is drawn and appended, with the real values for known-ahead rows; then the next 64.



So day length ahead can only steer the steps after the first 64.

Eleven time blocks for every row block keep attention cheap while related series still meet.

12 blocks (11 time, 1 row) · width 768 · 12 heads · patches of 64 · 24 Student-t components

In the notebooks: last 2,048 days or full monthly history · day length and calendar terms for D2 · 32 sample paths, median as the point